

Hierarchical Prediction and Feedforward Correction of Time-Varying Thermal Line-of-Sight Bias for Coarse Acquisition in Satellite Optical Communications

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01

Background

Before optical feedback is available,
scan-center offset determines coarse-acquisition search time

Before optical feedback

Search an uncertainty region
including **thermal LOS**, attitude, and
orbit errors

Illumination / eclipse / dissipation

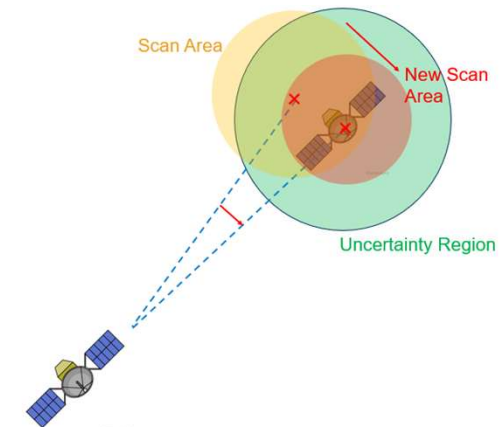
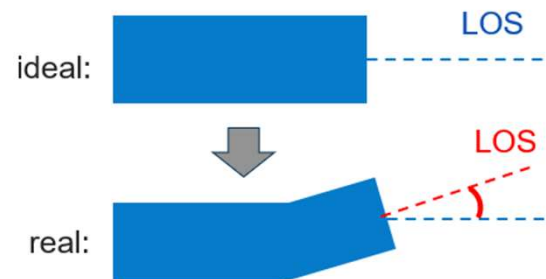
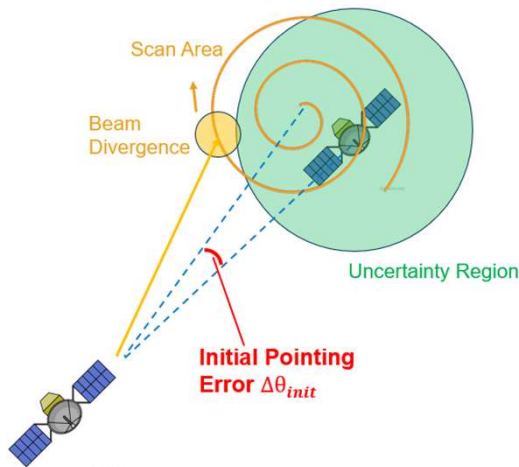
Temperature field → thermal expansion
→ spacecraft-bus distortion

Thermal LOS: 100 μ rad to 1 mrad

Our approach

Pre-acquisition predictive correction

Subtract **thermal LOS**
from the scan center
to reduce search time



Without modifying the spacecraft or LCT,
operationally remove only predictable thermal deformation from the initial LOS error.

Thermal LOS error can match or exceed orbit-prediction error—and cannot be optically separated before acquisition

Main contributors to scan-center LOS error during coarse acquisition

Error source	Approximate magnitude	Time variation
Thermal LOS	150 μ rad–1000 μ rad (this analysis)	Orbital period (~100 min in LEO)
Orbit prediction	Several hundred μ rad (TLE propagation vs. precise GPS ephemeris)	Orbital period (~100 min in LEO)
Attitude and calibrated alignment	Tens of μ rad	DC to broadband
Mechanical vibration	Several to tens of μ rad	High frequency (post-acquisition fine tracking)

* Thermal LOS and orbit-prediction error are from this analysis; other values are literature-based order-of-magnitude estimates.

Thermal LOS and orbit prediction share similar orbital periods
Frequency separation of post-acquisition residuals cannot isolate the thermal component



Do not separate thermal effects retrospectively from observations
Use temperature and operating state to subtract them before acquisition

Prior work on thermal-deformation LOS error prediction and this study

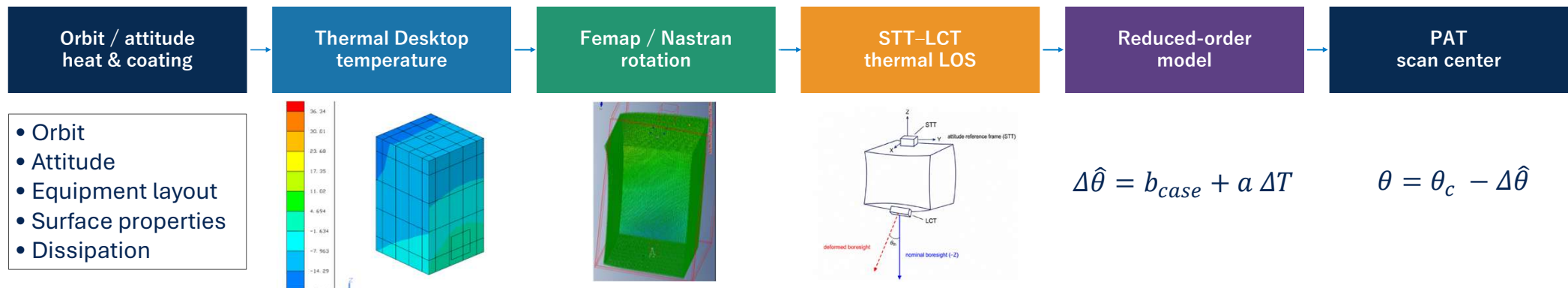
Field	Optical communications	Earth observation / optics	This study
Approach to thermal deformation	Reduce thermal deformation through structure, materials, and equipment placement Assess thermal pointing error by FEM	Predict and correct LOS variation from temperature, orbital phase, and observation geometry	Predict spacecraft-bus relative LOS from temperature and operating state
Main target	Optical terminals / spacecraft structures	Cameras / remote-sensing optical systems	STT-LCT relative LOS across the spacecraft bus
Evaluation metric	Thermal stability, pointing error, link establishment	LOS prediction error, image-pointing accuracy	Coarse-acquisition time and success rate
Remaining gap	Limited frameworks for predicting residual thermal LOS error during operations	Not linked to spacecraft-bus LOS error or coarse-acquisition performance	—

References:

- Optical comm.: Shi et al. 2023; Zhang et al. 2025
- EO/optics: Li et al. 2025; Hu et al. 2022; Turella et al. 2019/2021

From structural mitigation to instrument-specific LOS correction, this study develops a physics-based bus-level LOS model for coarse optical acquisition.

Research overview: from thermoelastic truth generation and model construction to evaluation



Ground model development

- Input orbit, attitude, and thermal environment
- Compute temperature fields and thermoelastic deformation with TD/Femap
- Generate reference STT-LCT relative-LOS time series
- Train and evaluate the reduced-order model

On-orbit correction

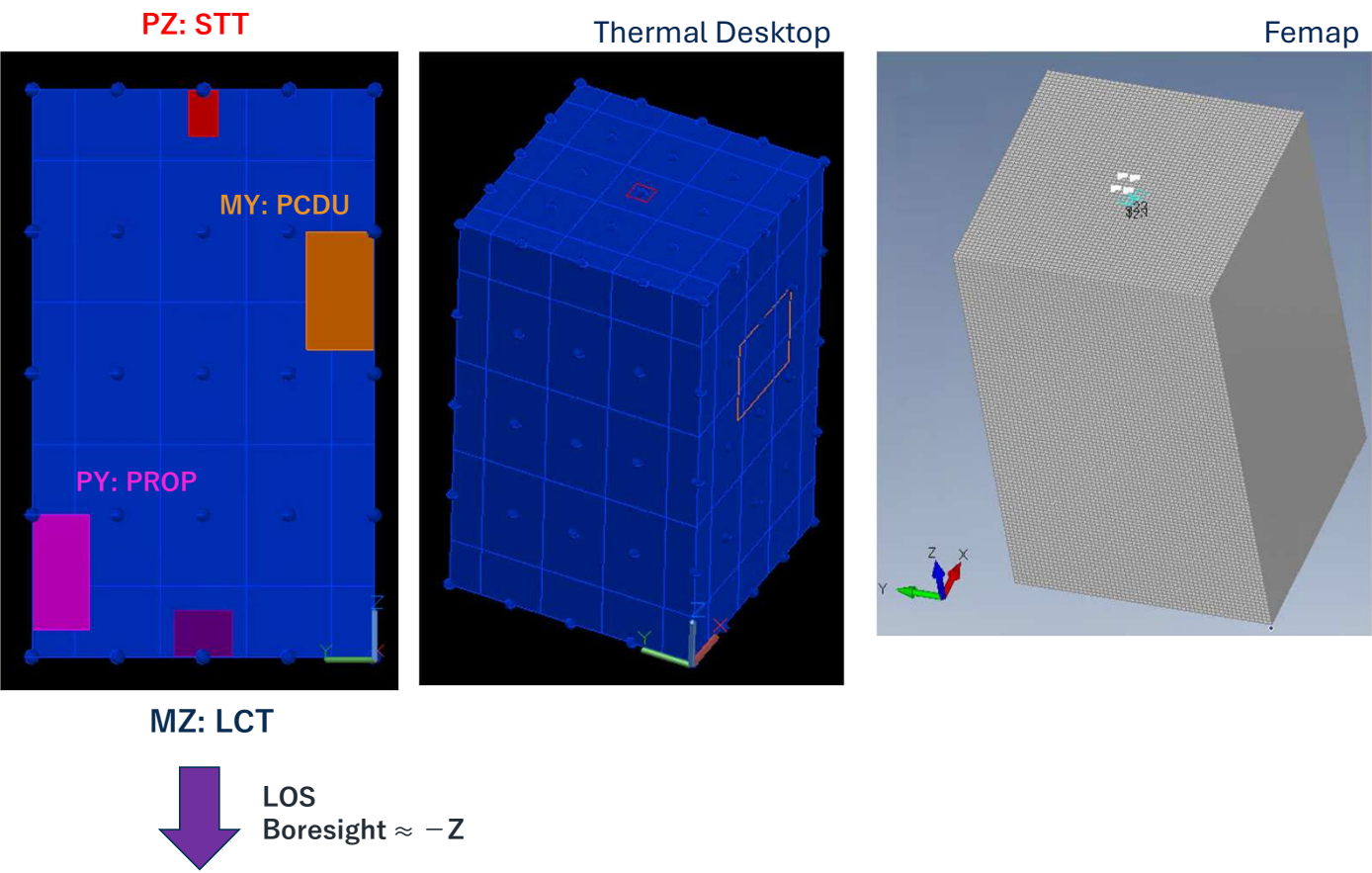
- Temperature / operating state
- Predict thermal LOS
- Correct the scan center
- Include nonthermal errors
- Residual update

- TD/Femap runs only on the ground; only the reduced-order model runs onboard
- Evaluate LOS error, acquisition success, and acquisition time

02

Thermoelastic analysis

We evaluate a box-structure LEO spacecraft with the STT and LCT on different panels



Main analysis-model settings

Item	Setting
Bus dimensions	590 × 600 × 990 mm
Panels	10-mm-thick shell elements
Material	A5052
LCT	Center of MZ panel
STT	Center of PZ panel

Not a specific-spacecraft design review.
Fixed structure; sun face and dissipation vary.

Thermal Desktop generates temperature-field time series from orbit, attitude, surface, and dissipation inputs

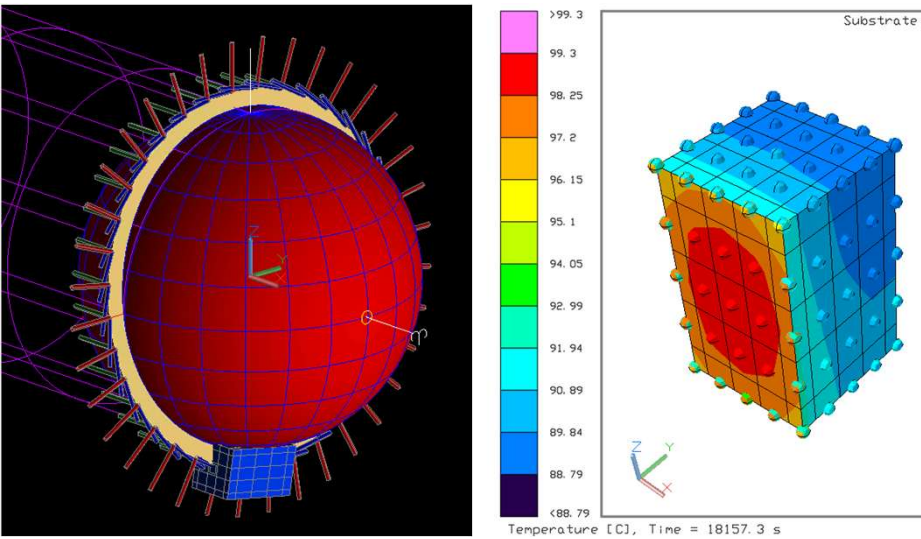
Main thermal-analysis inputs

Input	Setting
Orbit	Sun-synchronous, LTAN06 / 800 km / COLD
Period	≈6050 s (≈101 min)
Analysis duration	≈18,157 s (3 orbits)
Sampling	Every ≈60.5 s, 301 points
Illumination	Eclipse / continuous illumination
Sun face	MX / MY / PX / PY
Number of nodes	125

Output: panel temperature field T(t), computed on the ground.

Environmental conditions varied by case

- Internal-unit power: ON/OFF
- Surface: $\alpha = \varepsilon = 0.5$ / Black / Alodine
- Orbit: LTAN18 / 693 km

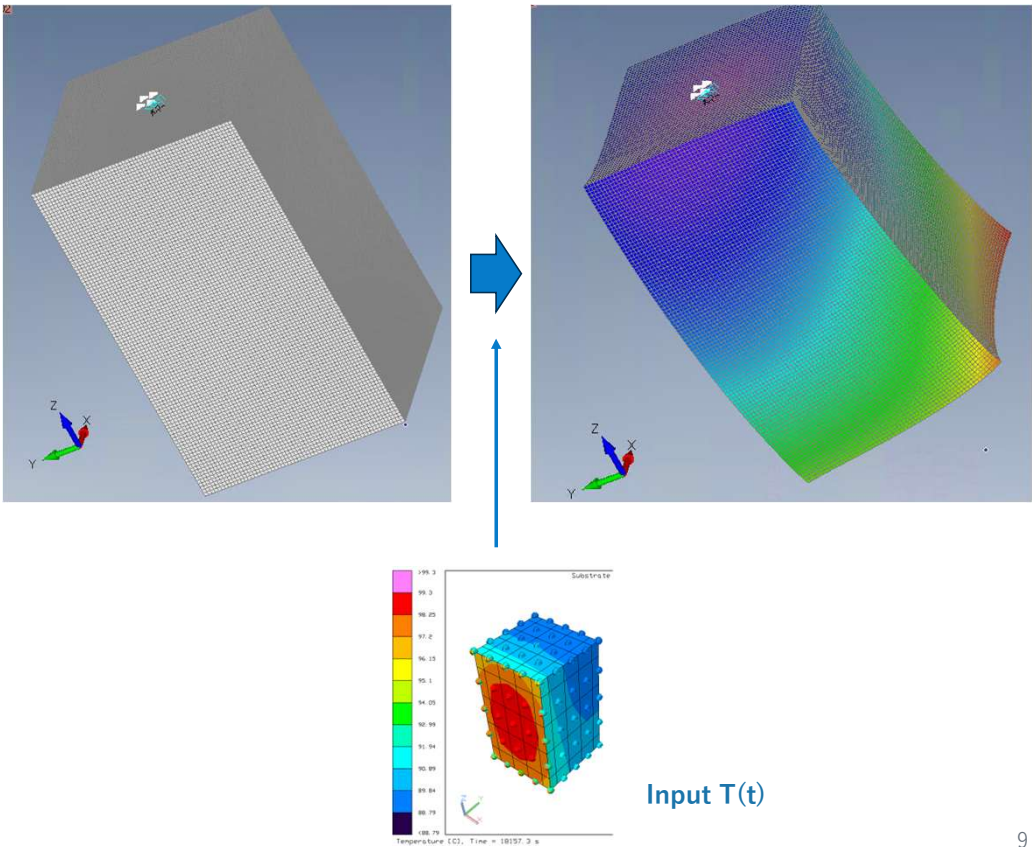


The temperature field is mapped into Femap to obtain STT and LCT translations and rotations

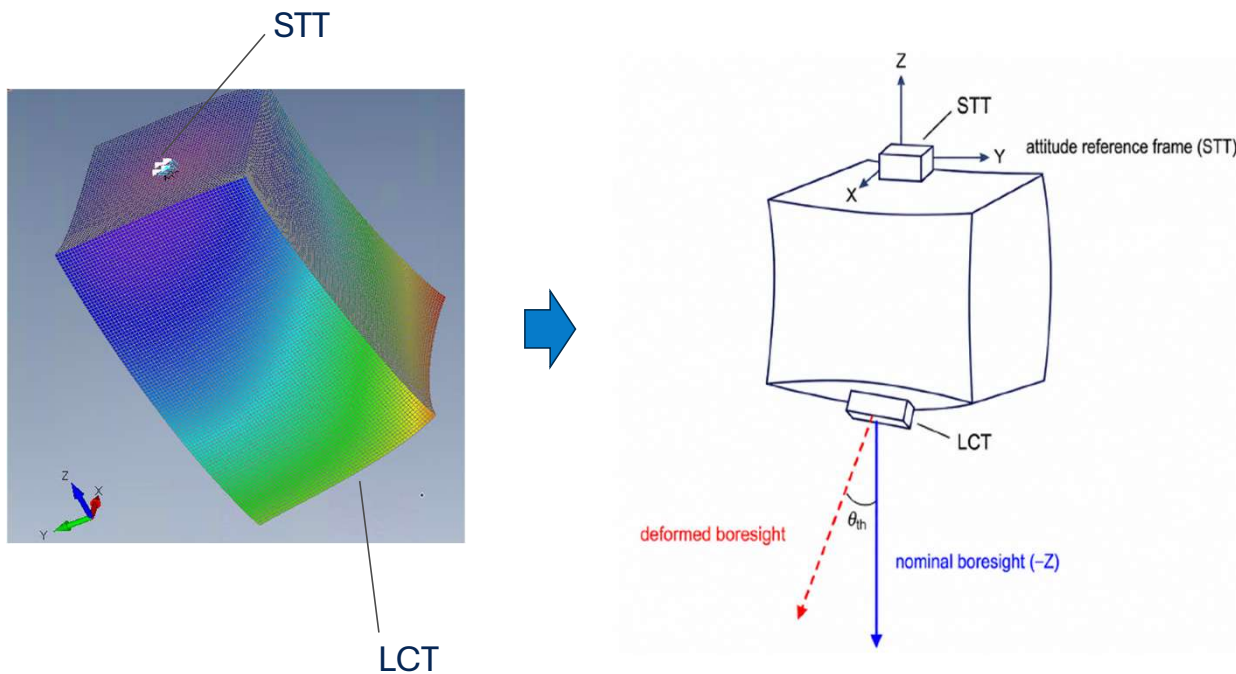
Main structural-analysis settings

Structural-analysis setting	Value
Material	Aluminum A5052
Young's modulus	70.327 GPa
Poisson's ratio	0.33
Coefficient of thermal expansion	$2.376 \times 10^{-5} / \text{K}$
Density	2685 kg/m ³
Reference temperature	23.9 °C
Constraint	Small region near the STT

Thermal Desktop and Femap generate training data on the ground.



For a far-field link, the relevant quantity is rotation of the LCT optical axis relative to the STT



Far-field LOS error = STT-relative LCT optical-axis rotation, not translation

LOS-error definition

- LCT optical-axis rotation – STT reference rotation
- STT thermal rotation is absorbed into attitude
- Directly gives far-field scan-center error
- Excludes translation-induced tilt

$$\theta_{thermal,true} = \theta_{LCT,rot} - \theta_{STT,rot}$$

$$\theta_{thermal,true} = [\theta_x, \theta_y]$$

The structure and LOS definition are fixed; only sun face, dissipation, surface, and orbit vary

Case parameters

Varied factor	Cases	Purpose
Sun face	MX / MY / PX / PY	Dominant axis, sign, and magnitude
Internal dissipation	STT/LCT only / +PROP / +PCDU / all units (one half-power case)	Variation in mean bias across cases
Surface properties	Baseline ($\alpha = \varepsilon = 0.5$) / Black / Alodine	Variation amplitude and residual floor
Orbit / thermal environment	Eclipse / continuous illumination; LTAN06 / LTAN18	Applicability across thermal and orbit conditions

Fixed parameters: structure and dimensions, STT/LCT placement, material, and LOS definition

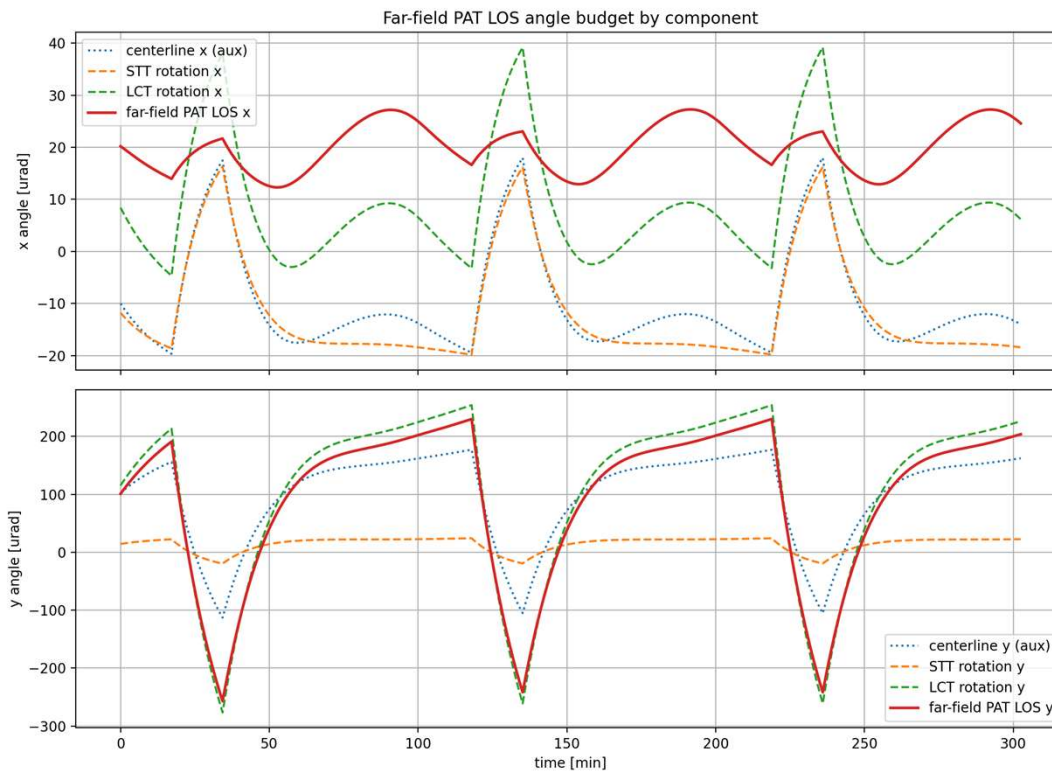
Current analysis: 21 cases.

Across-condition evaluation is limited to the same spacecraft bus; coefficients are not transferred unchanged to another structure.

PZ/MZ sun-face cases are not included;
they host the STT/LCT and are outside this model's scope.

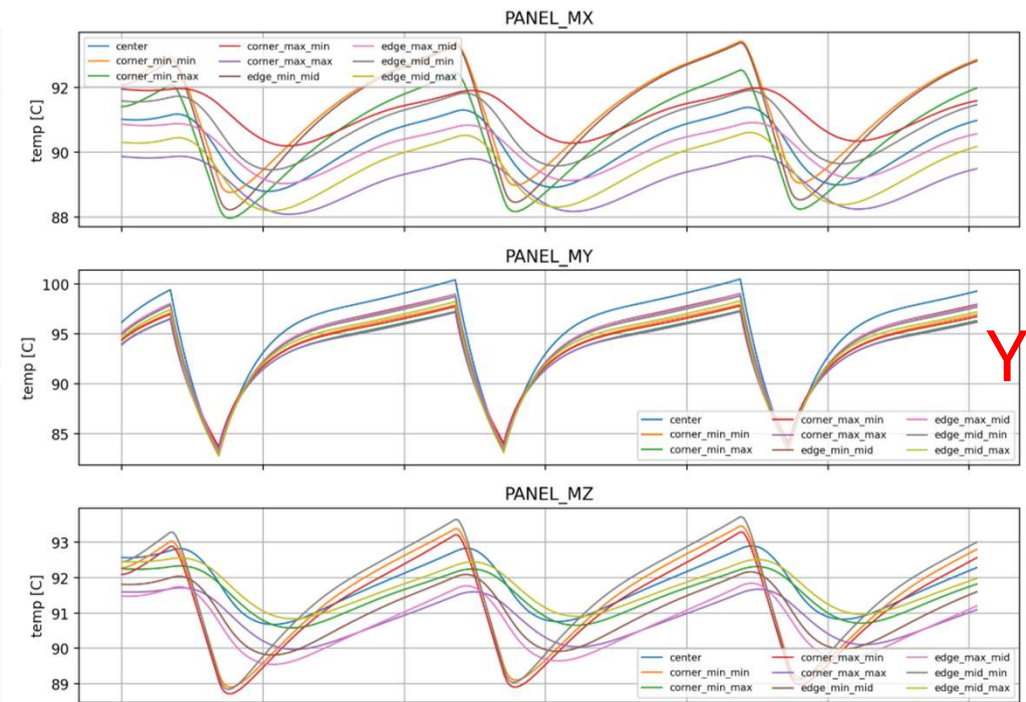
Results: thermal LOS varies at the orbital period; the dominant axis and mean offset follow the sun face

Y



Case 04 (MY sun face): far-field LOS

- Orbital-period thermal LOS; dominant axis: MY/PY $\rightarrow$ y, MX/PX $\rightarrow$ x
- Dominant-axis variation follows sun-facing-opposite-panel center ΔT
- Raw RMS: 150–1280 μ rad; surface sets amplitude/floor, dissipation shifts the case mean



Panel temperatures

03

Reduced-order model

Reduced-order model selection

Model requirements

- No onboard TD/Femap
- Few temperatures and operating flags
- Small, fixed coefficient set
- Output: scan-center correction angle

Candidate model	Strengths / limitations
Constant bias	Lightweight / cannot represent within-orbit variation
Fourier series in orbital phase	Represents periodicity / must be rebuilt when orbit or conditions change
Multiple mounting-point temperatures	Flexible / collinearity with ΔT destabilizes coefficients
Sun-face temperature T	Physically interpretable / does not fully track variation
Sun-face temperature difference ΔT	Physically interpretable / cannot explain mean offsets across cases
Hierarchical ΔT	Separates within-orbit variation from means across cases

→ Adopt a hierarchical model that separately predicts thermal-LOS variation from sun-face-opposite-panel ΔT and thermal-LOS bias from the mean offset across cases

Onboard prediction separates time-varying and bias terms using ΔT and operating flags

$$\theta_{dom}(t) \approx a_{sunface} \cdot \Delta T(t) + b_{case}$$

$$\Delta T(t) = T_{sunface}(t) - T_{opposite}(t): \text{sun face} - \text{opposite panel}$$

Assumption: onboard panel-center thermocouples

$$b_{case} \approx b_{0,sunface} + c_{PROP} I_{PROP} + c_{PCDU} I_{PCDU}$$

I_{PROP}, I_{PCDU} : unit power state (ON/OFF)

Dominant-axis selection

Prediction axis from sun face
MX/PX $\rightarrow$ x; MY/PY $\rightarrow$ y
Non-dominant axis: bias only
MX/PX bias $\approx -600 \mu\text{rad}$;
zeroing it degrades acquisition

$a_{sunface}$

Sensitivity $a(\text{sun})$
Shared by sun face
+30.6, +28.6, -28.1, -28.7 $\mu\text{rad/K}$

b_{case}

Case-dependent DC offset
From sun face and ON/OFF flags
Decomposed by the equation above

Panel ΔT : time variation · DC offset: cross-case differences

16 coefficients; specific to this structure, placement, and LOS definition

The test case is excluded from all coefficient estimates to evaluate prediction under unseen conditions

① Fit each case

Use the first orbit of each case
to estimate its sensitivity a
and mean b



② Share within each sun face

Median a within each sun face
becomes the shared sensitivity

Fit mean b across cases from
sun face and dissipation flags



③ Test an unseen case

nested leave-one-case-out

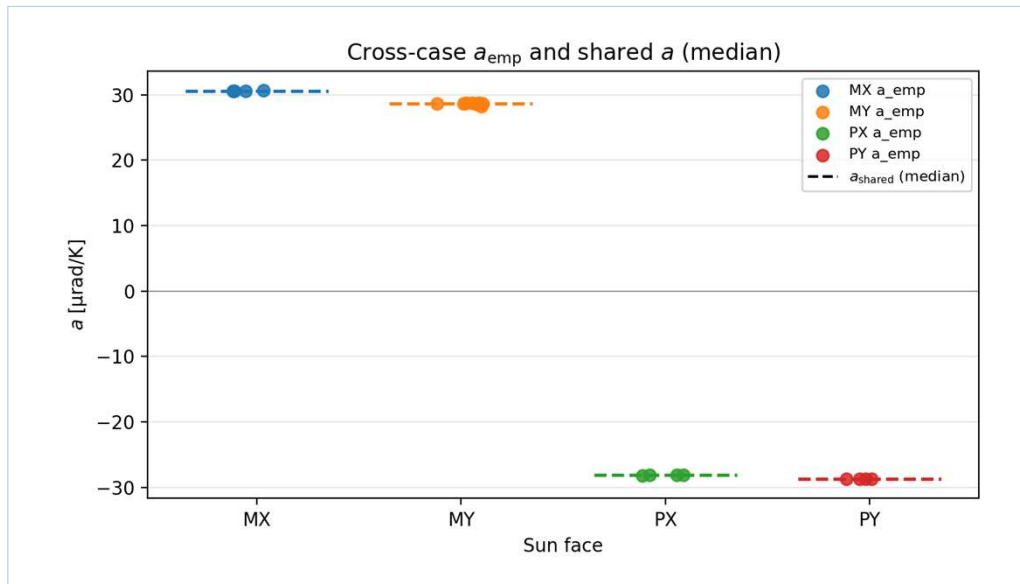
Exclude the test case from shared sensitivity
and both-axis mean models

Evaluate its next two orbits

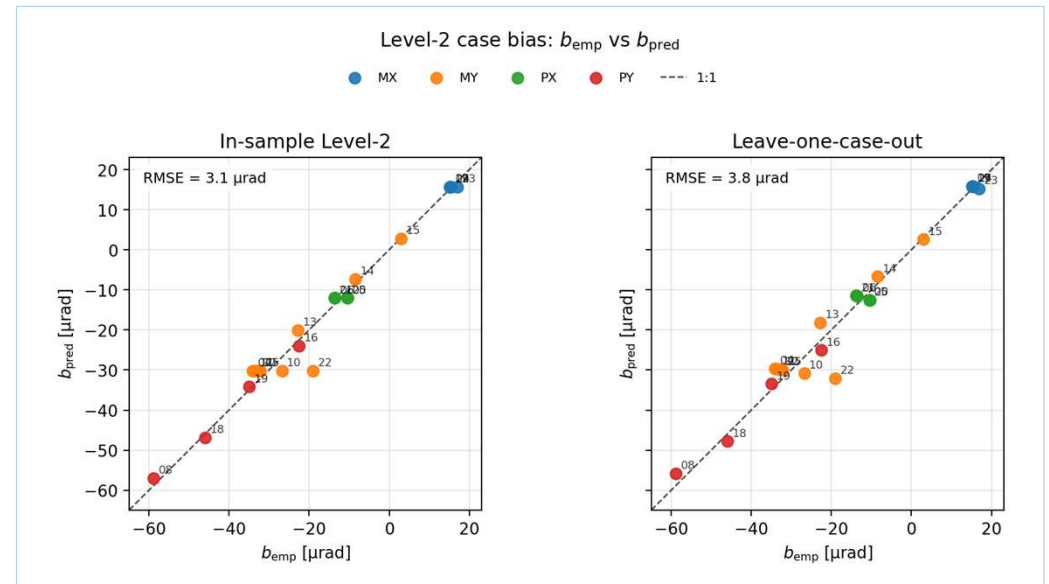
Evaluate how well fixed shared coefficients predict unseen operating conditions

Across-condition evaluation within the same structure and LOS definition

For unseen cases, median dominant-axis error is $4.9 \mu\text{rad}$ —one to two orders below the raw $615 \mu\text{rad}$



Sun-face sensitivities a (case-specific and shared)



Case bias b : empirical versus predicted

$615 \mu\text{rad}$

Median raw dominant-axis LOS RMS

$4.9 \mu\text{rad}$

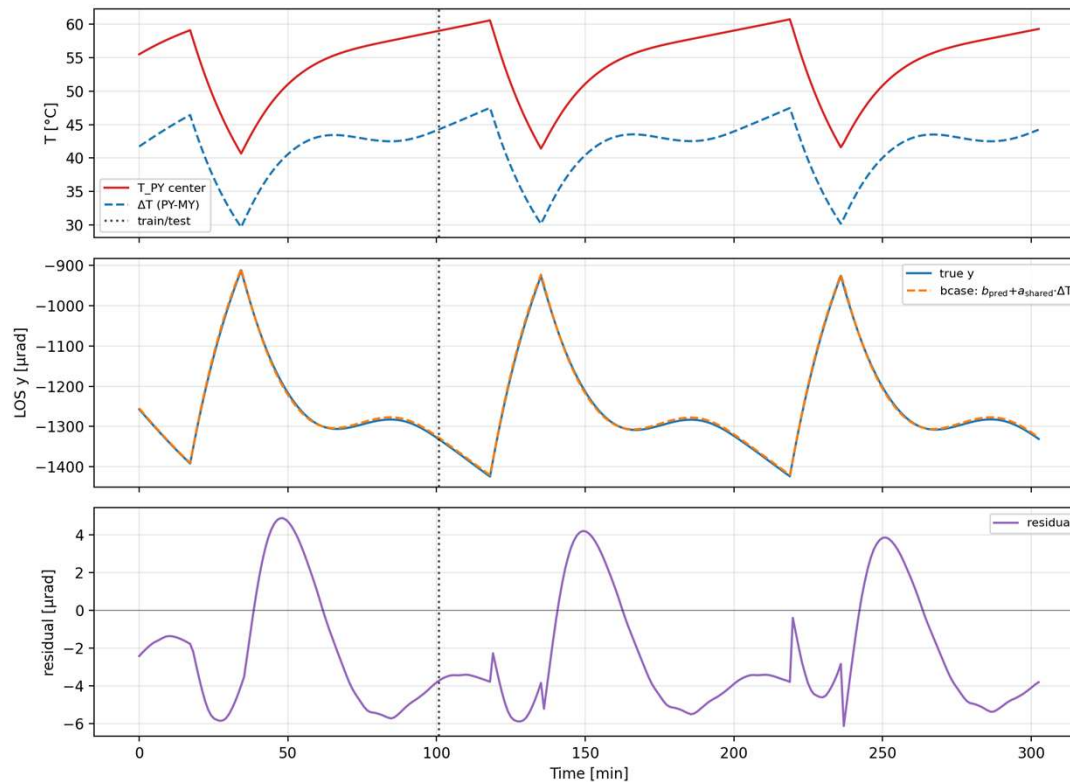
Median test RMSE on unseen cases (mean 5.5)

$3.8 \mu\text{rad}$

LOO RMSE of case-mean DC bias

Shared sensitivity: MX +30.6 / MY +28.6 / PX -28.1 / PY -28.7 $\mu\text{rad/K}$

Even for Case 08, with one of the largest thermal LOS errors, 1250 μrad falls to 3.9 μrad



Case 08 (PY sun face, all units dissipating)

1250 μrad

Raw dominant-axis RMS

3.9 μrad

Test RMSE for an unseen case

First scan point is sufficient

Corrected residual is well below
the 150 μrad detection radius

ΔT follows the within-orbit variation

04

Coarse-acquisition evaluation

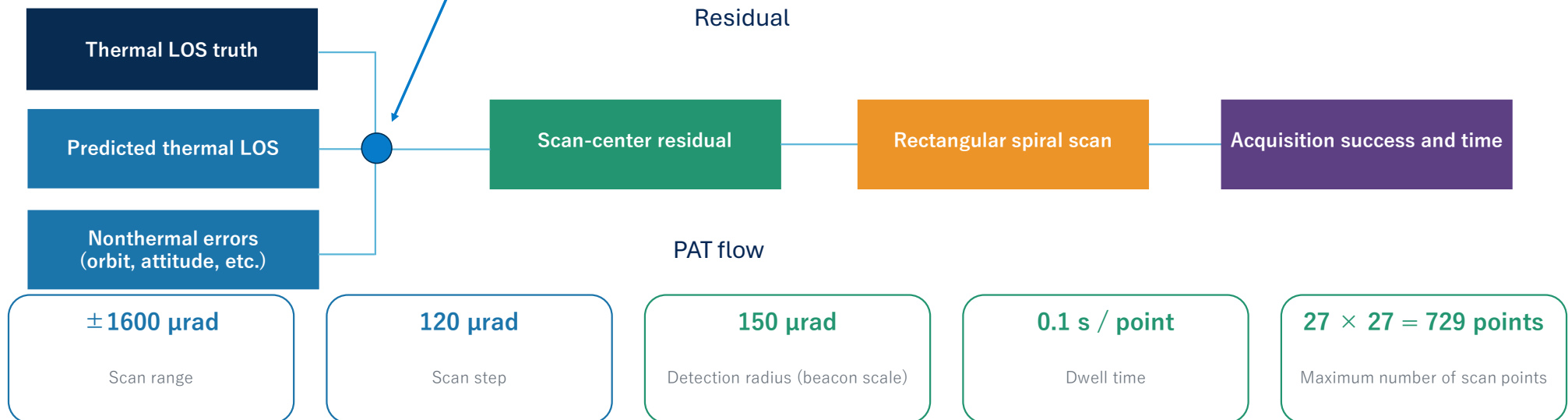
Acquire when the scan-corrected target enters the detection circle

$$e_{scan}(t) = e_{nonthermal}(t) + \theta_{thermal,true}(t) - \hat{\theta}_{thermal}(t)$$

Scan-center error Orbit, attitude, and other nonthermal errors Thermal LOS truth Predicted thermal LOS

Residual

PAT flow



No coverage holes: half the grid diagonal $\approx 85 \mu\text{rad} < 150 \mu\text{rad}$ detection radius

Neglects inter-point slew, settling, and stochastic detection. Times are comparative under these scan conditions.

Orbit prediction, alignment, attitude, and drift are added to the thermal error

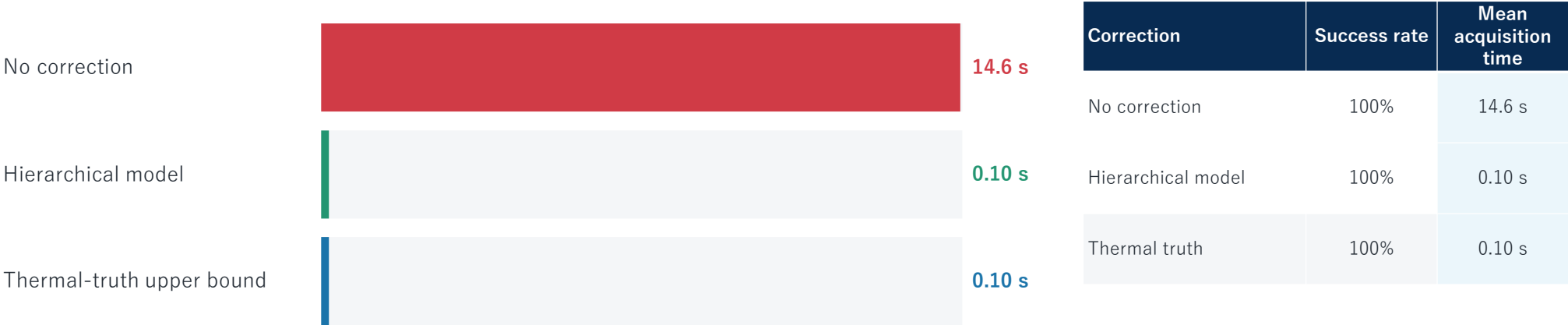
$$e_{nonthermal} = e_{orbit} + e_{alignment} + e_{attitude} + e_{drift}$$

Component	Construction	Time behavior	Magnitude
Orbit prediction	Propagate the latest Sentinel-1 TLE; project error from precise POEORB onto the link transverse plane	Orbital period	Mean 280–700 μrad
Alignment	Constant two-axis residual per case	DC	50 μrad/axis, 1 σ
Attitude	Gaussian random draw at each sample	Simplified broadband	50 μrad/axis, 1 σ
Drift	Sinusoid	900 s period	30 μrad amplitude

One random seed per case; statistical spread is not evaluated.

This is not a spacecraft-specific error budget; it represents a generic GNSS-denied small-LEO scenario.

With thermal LOS error only, mean acquisition time drops from 14.6 to 0.10 s, matching thermal-truth correction



9.3 μ rad
Mean thermal residual after correction

The residual is far below the 150 μ rad detection radius, so acquisition occurs in one dwell.
Mean over 14 COLD baseline-surface cases; the main result includes nonthermal error on the next slide.

With nonthermal LOS error, the mean drops from 19.2 to 5.45 s; the residual is governed by nonthermal error



Metric	No correction	Hierarchical model (feedforward)
Success rate	98%	100%
Mean acquisition time	19.2 s	5.45 s
Mean initial error after correction	—	448 μrad (almost entirely nonthermal)
PY sun face (thermal ≈1.2 mrad)	37.3–39.5 s / 88–97%	1.3–1.8 s / 100%

Predicting and removing thermal error shifts the coarse-acquisition limit from thermal to nonthermal error

05

Summary

Limitations of this study

Validation scope

Numerical evaluation: same structure and LOS definition

Model applicability

No ground test; flight relevance of the analysis model and sensitivities remains unvalidated

LOS extraction

Representative-node rotations; no interface rigid-body fit or internal STT/LCT deformation

Acquisition evaluation

One nonthermal seed; simplified scan dynamics

Removing predictable thermal error before acquisition leaves a search governed by nonthermal error

4.9 μ rad

Median corrected dominant-axis RMSE
(raw median: 615 μ rad)

14.6 $\rightarrow$ 0.10 s

Mean acquisition time, thermal only

19.2 $\rightarrow$ 5.45 s

Mean acquisition time with nonthermal error
Success rate: 98% $\rightarrow$ 100%

Results: For the evaluated box bus, the method can shorten pre-feedback link establishment.

Contributions: STT-LCT relative LOS, cross-condition coefficient sharing, and connection to coarse-acquisition performance.

Future Work: ground-test validation, coefficient re-identification for real spacecraft,
and onboard adaptation using post-acquisition residuals

Oral presentation at ICSO 2026 on October 15

Thank you for your attention